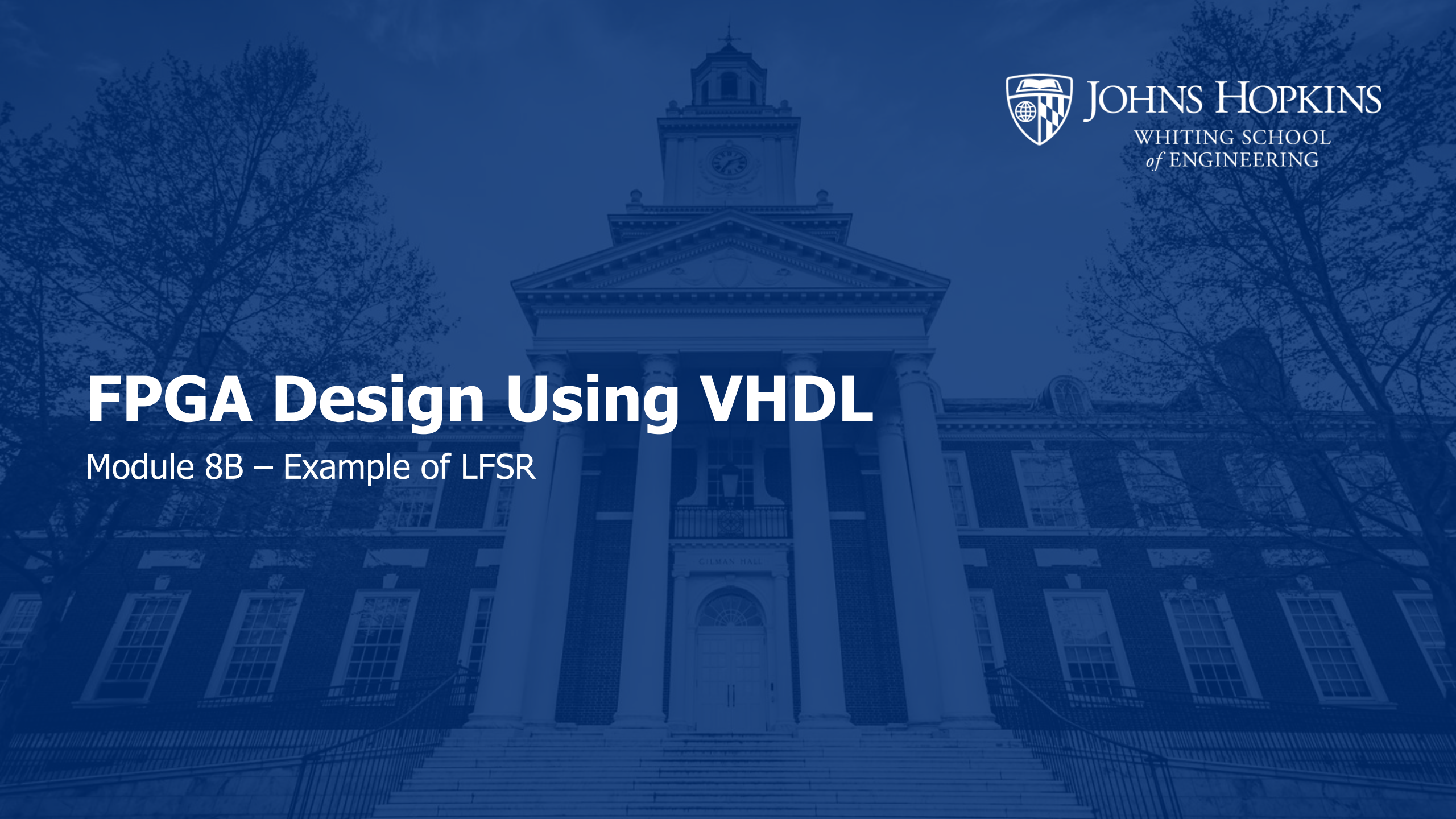


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FPGA Design Using VHDL

Module 8B – Example of LFSR

VHDL Example

- 4 bit maximal length LFSR
- Maximal length polynomial: x^3+1
- Initial state (seed) "1111"
- 15 non-zero states
- Repeats every 15 clock cycles
- Write VHDL using concatenation
- Write the process only
  - Assume I/O are defined

VHDL code for LFSR Design

```
library ieee;
use ieee.std_logic_1164.all;
use work.all;

ENTITY LFSR IS
  port(
                clk_50   : in  std_logic;
                reset    : in  std_logic;
                seed     : in  std_logic_vector(3 downto 0);
                dout     : out std_logic_vector(3 downto 0)
      );
END ENTITY;

ARCHITECTURE behav of LFSR IS

signal dout_tmp : std_logic_vector(3 downto 0);

BEGIN

dout <= dout_tmp;

process(clk_50, reset)
begin
    if(reset = '1') then
                dout_tmp <= seed;
    elsif(rising_edge(clk_50)) then
                dout_tmp <=  dout_tmp(2 downto 0) & (dout_tmp(3) xor dout_tmp(0));
        end if;
end process;

END ARCHITECTURE behav;
```

Testbench to test LFSR

```
library IEEE;
use ieee.std_logic_1164.all;
use IEEE.std_logic_textio.all;
use work.all;


library STD;
use std.textio.all;

entity LFSR_tb is
end LFSR_tb;

architecture testbench of LFSR_tb is

signal clock : std_logic := '0';
signal reset : std_logic := '0';
signal seed : std_logic_vector(3 downto 0);
signal dout : std_logic_vector(3 downto 0);

begin

→ LFSR_inst: entity LFSR port map (
      clk_50 => clock,
      reset => reset,
      seed => seed,
      dout => dout);
```

```
process
begin
     reset <= '0', '1' after 20 ns, '0'
              after 100 ns;
     seed <= "1111";
     wait;
end process;

process
begin
     clock <= not clock;
     wait for 10 ns;
     clock <= not clock;
     wait for 10 ns;
end process;

process(clock)
   file LFSRfp : text open write_mode is
                 "LFSRout.txt";
      variable line_out : line;
      begin
         if(rising_edge(clock)) then
         → hwrite(line_out, dout);
         → writeline(LFSRfp, line_out);
         end if;
   end process;
end testbench;
```

LFSR Output Files

